



Hurricane Damage Detection with Deep Learning

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1. Introduction

The increasing societal exposure to natural hazards necessitates rapid and scalable methods for post-disaster response. Hurricanes, exemplified by the Category 4 Hurricane Harvey, which struck Texas in 2017, inflict widespread destruction upon infrastructure and housing. This single event resulted in at least 68 fatalities and approximately \$125 billion in damages, highlighting the immense economic and social consequences of such catastrophes [1]. Following these events, a rapid damage assessment is critical for prioritizing rescue operations, allocating resources, and estimating financial losses. Traditional assessment methods relied on labor-intensive manual surveys and the visual interpretation of aerial imagery. These approaches are time-consuming, prone to human error, and often hazardous for ground crews in the immediate aftermath of a storm [2]. While satellite imagery offers a comprehensive, near-real-time view of affected zones, the manual analysis of thousands of images creates a bottleneck that delays decision-making [3]. Consequently, automated systems capable of processing remote sensing data to identify structural damage efficiently have become a research objective. Early automated solutions employed object-based image analysis and shallow machine-learning algorithms, such as Support Vector Machines. However, these methods were limited by their reliance on hand-crafted features, which struggled to capture the complex spatial hierarchies required to distinguish subtle structural failures [4].

1.1 Deep Learning on the Hurricane Harvey Dataset

Research on automated damage detection using the Hurricane Harvey dataset [5] has shifted from custom, lightweight Convolutional Neural Networks (CNNs) to regularized transfer learning and multi-scale architectures. Cao and Choe [6] evaluated a custom CNN comprising four convolutional layers, max-pooling, and dropout against a *VGG16* transfer-learning baseline. Their custom architecture outperformed the pre-trained model, achieving an accuracy of 97.08% (97.29% on a balanced set), an F1-score of 0.9723, and an AUC of 0.996. Kaur et al. published two studies that explore the trade-offs between architectural depth and computational efficiency. In their first iteration [2], they proposed a 14-layer custom CNN consisting of five convolutional layers, five pooling layers, a flattening layer, and two dense layers. This model achieved an accuracy of 92.91% and a sensitivity of 93.34%. Subsequently, Kaur et al. [7] optimized this approach using a 15-layer architecture, incorporating an additional dropout layer. This refinement improved validation accuracy to 95.19% and precision to 0.97. Simultaneously, Kaur et al. [8] evaluated transfer learning backbones, including *VGG16*, *MobileNetV2*, *InceptionV3*, and *DenseNet121*. *VGG16* proved most effective, but while achieving high recall (0.95), the overall accuracy remained between 75% and 78%. Later in 2022, Bao [9] inverted the earlier findings regarding custom versus pre-trained models. By applying regularization with data augmentation, batch normalization, and weight decay to a *VGG16* backbone, Bao achieved an accuracy of 99.15%. Concurrently, Islam et al. [10] proposed a *Compact Convolutional Transformer* to address the limitations of pure CNNs. Their model combined convolutional layers for local feature extraction via tokenization with three transformer layers for processing global context. This hybrid architecture achieved an accuracy of 98.79% and a recall of 99.05% while using only 0.57 million parameters. Furthermore, they integrated *Local Interpretable Model-agnostic Explanations* to validate that the model correctly focused on building structures rather than background noise. Following this, Jawahar et al. [11] introduced *TRS-Net* to address multi-scale feature extraction. Utilizing depth-wise convolutions with varying dilation rates, *TRS-Net* achieved 98.57% accuracy and a perfect sensitivity of 1.0. This metric is operationally critical, as it indicates zero false negatives for the reported test split, but it does not guarantee the same property under geographic or temporal shifts. Most recently, Wang et al. [12] modified the *VGG16* architecture by replacing dense classification layers with *Global Average Pooling* to reduce parameter complexity. This method achieved an accuracy of 97%, which is lower than the performance reported by Bao, Islam et al., and Jawahar et al., suggesting that removing dense layers may compromise the feature discrimination in this case.

Table 1.1: Reported performance metrics on the Hurricane Harvey dataset. The data illustrate a progression from lightweight custom models to high-accuracy regularized transfer learning and hybrid architectures, followed by recent trade-offs for sensitivity and parameter efficiency.

Study	Architecture Details	Accuracy (%)	Sensitivity	F1-Score
Cao and Choe (2020) [6]	Custom CNN (4 Conv layers, Dropout)	97.08	N/A	0.97
Kaur et al. (2022) [2]	Custom CNN (14 Layers)	92.91	0.93	0.93
Kaur et al. (2022) [7]	Custom CNN (15 Layers)	95.19	0.95	0.96
Kaur et al. (2022) [8]	VGG16 (Transfer Learning)	78.00	0.95	0.83
Bao (2022) [9]	VGG16	99.15	N/A	High
Islam et al. (2022) [10]	Compact Conv. Transformer	98.79	0.99	0.99
Jawahar et al. (2023) [11]	TRS-Net (Depth-wise Convolution)	98.57	1.00	0.99
Wang et al. (2024) [12]	VGG16 + Global Average Pooling	97.00	N/A	N/A

1.2 Semantic Categorization and Multimodal Integration

While binary classification remains the primary testbed, the field is expanding toward higher-level semantic tasks and multimodal integration to address the complexity of damage typologies and sensor limitations. Xiao and Mostafavi [13] introduced *DamageCAT* to transition from binary labels to granular damage typologies, specifically distinguishing between partial/total roof damage and partial/total structural collapse. Utilizing a hierarchical U-Net-based transformer with multi-scale difference blocks on the BD-TypoSAT dataset from Hurricane Ida, the model achieved an overall Intersection over Union of 0.737 and an F1-score of 0.846. This work evaluated transferability across Hurricanes Harvey, Florence, and Michael, identifying generalization failure modes in fine-grained classification tasks.

To mitigate the limitations of optical sensors under cloud cover and during storm surges, Ho and Mostafavi [14] developed *Flood-DamageSense*. This framework fuses pre- and post-event SAR/InSAR data with optical basemaps and a historical flood risk layer that encodes long-term exposure probabilities to guide detection in areas with minimal spectral change. Built on a multimodal *VMamba* [15] backbone, the model utilizes multitask learning to jointly predict graded damage states, floodwater extent, and building footprints.

1.3 Integrity and Reliability

Despite the quantitative success reported in the literature, where accuracies on the Harvey dataset exceed 97% (e.g., [9, 11, 10]), a gap remains regarding the reliability of these metrics and the integrity of the underlying dataset. The observation that diverse architectures achieve near-perfect performance could indicate that the dataset may no longer effectively discriminate among model capabilities. Furthermore, high accuracy does not imply that a model provides trustworthy probability estimates. In emergency response, a system that is 99% accurate but overconfident in its inevitable errors poses significant risks. Notably, none of the reviewed studies (see Table 1.1) reported calibration metrics such as Expected Calibration Error (ECE) or Brier score. The concept of calibration has been studied in general deep learning [16, 17]. Metrics such as ECE [18] and techniques such as Binning [19] or Spline-based calibration [20] provide tools for auditing model trustworthiness.

This project challenges the validity of the reported metrics on the Hurricane Harvey dataset, and the codebase is maintained on GitHub [21]. It is hypothesized that the feature space of the Harvey dataset contains exploitable artifacts, known as “shortcut learning” [22]. If models rely on spatial or temporal confounding variables rather than structural damage features, the reported accuracies are over-optimistic estimates of operational performance. Consequently, this project evaluates the model behavior under both the original dataset and a resampled dataset, focusing on two potential issues: (1) data leakage induced by an erroneous splitting strategy, and (2) confidence calibration and reliability assessment.

2. Exploratory Data Analysis

This chapter characterizes the dataset to identify properties affecting model performance. The analysis reveals leakage risks, which provide the rationale for a resampling strategy.

2.1 Dataset Inventory and File Semantics

The dataset comprises RGB image tiles for binary classification between *damage* (flooded, structural failure) and *no_damage*. The imagery covers the Greater Houston area following Hurricane Harvey in 2017 [1] (see Fig. 2.1 for spatial distribution). The data include sub-meter-resolution RGB sensor imagery that has been orthorectified and pansharpener. The ground truth labels were generated via the Tomnod crowd-sourcing platform. Positive samples (*damage*) were identified by volunteers from post-event imagery, and negative samples (*no_damage*) were generated by cropping from pre-event imagery at the corresponding geographic coordinates [6].

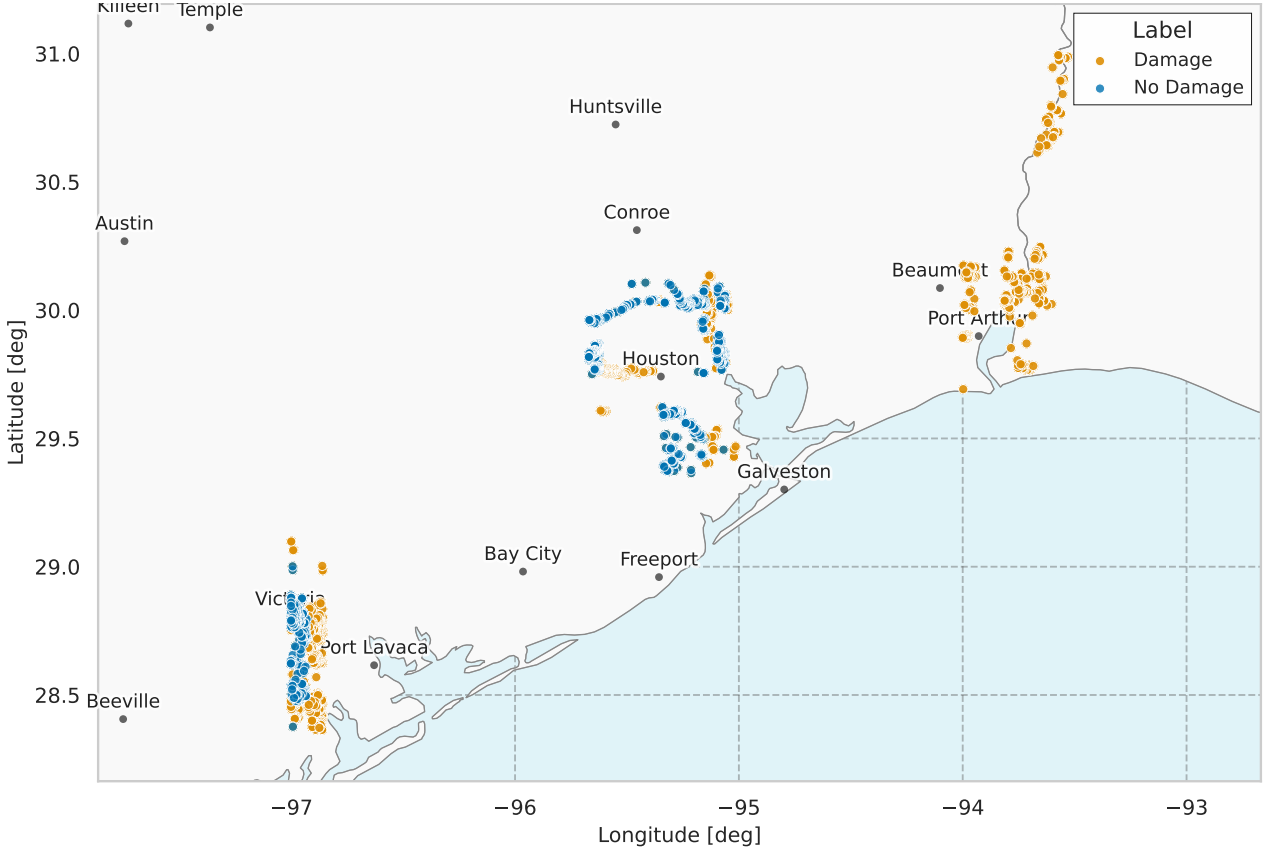


Figure 2.1: Geospatial distribution of the dataset across the Greater Houston area. Orange markers indicate *damage* samples (derived from post-event imagery), while blue markers indicate *no_damage* samples (derived from pre-event imagery).

The dataset comprises 23'000 images (15'000 *damage*, 8'000 *no_damage*) partitioned into balanced *validation* and *test* sets (1'000 images per class each) and an imbalanced *train* set (13'000 *damage* vs. 6'000 *no_damage*). Although Cao and Choe used 128×128 crops resized to 150×150 [6], the tiles are consistently 256×256 pixels. Integrity checks (*PIL.Image.verify()*) confirmed no file corruption. Geospatial coordinates were parsed directly from filenames ($\langle lon \rangle _ \langle lat \rangle$) to serve as proxies for tile locations. While some sensor parameters (GeoEye-1, 46 cm resolution) and extraction settings are documented [6], precise acquisition timestamps or other metadata for individual tiles are unavailable.

2.2 Methodological Critiques and Potential Data Leakage

Cao and Choe [6] construct the dataset by exclusively mapping *damage* labels to post-event imagery and *no_damage* labels to pre-event imagery at identical coordinates. This design introduces a confound between the target class and the temporal acquisition domain. The resulting label-aligned domain shift reduces the classification task to domain discrimination, enabling shortcut learning [22] via global photometric artifacts (e.g., solar angle, atmospheric haze, sensor noise) rather than local morphological analysis. This reliance on spurious correlations precipitates three critical validity threats:

1. **Metric Inflation via Domain Separation:** Reported accuracies ($> 97\%$) could quantify the model’s ability to distinguish temporal domains (Pre- vs. Post-event) rather than its sensitivity to structural damage.
2. **Spectral Overfitting:** The model risks functioning as a low-frequency spectral discriminator, optimizing for global color histogram shifts inherent to the temporal gap rather than high-frequency spatial features of debris or rubble.
3. **Operational Generalization Failure:** A model conditioned on this distribution could fail in “terra incognita” [23], specifically yielding false positives on *no_damage* post-event structures that share the atmospheric and lighting characteristics of the training set’s *damage* class.

The following subsections delineate the specific sources of this potential data leakage.

2.2.1 Data Integrity: Induced Spatial Bias via Temporal Pairing

While the study claims to associate each coordinate with a “unique” image to resolve overlapping strips (Section 3.3, p. 8, [6]), the negative sampling strategy explicitly generates *no_damage* samples by extracting pre-hurricane imagery at the identical coordinates of the “Flooded/Damaged” targets (Section 3.2, p. 7, [6]). Consequently, the dataset is intentionally constructed of paired pre- and post-event views of the same physical structures. This procedure violates the assumption of sample independence by treating temporal snapshots of the same subject as distinct instances. As a result, the target variable (structural state) is confounded with subject identity, prompting the model to distinguish classes based on temporal variability (e.g., seasonal lighting or vegetation changes) rather than on generalized morphological evidence of damage.

2.2.2 Feature Learning: Confounding via Background Collinearity

The retention of image pairs at identical geographic coordinates (Section 3.2, p. 7, [6]) introduces high collinearity in the feature space. In the absence of total landscape destruction, background features (e.g., road geometries, vegetation, driveway layouts) remain static across opposing class labels. Given the texture bias of standard CNN architectures [23], the high signal-to-noise ratio between static background and local-damage features encourages “shortcut learning” [22]. The network minimizes loss by attending to global radiometric statistics or high-frequency artifacts (e.g., lighting shifts, seasonal vegetation changes) rather than morphological evidence of destruction. The resulting decision boundary is thus defined by nuisance variables that are required to distinguish identical spatial arrangements, rendering the model brittle to environmental variation.

2.2.3 Validation Strategy: Spatial Leakage and Inflation

The study ignores spatial autocorrelation (Tobler’s First Law of Geography). Cao and Choe acknowledge that “[...] *the same coordinate may appear in both the training and test sets*” (Section 3.3, p. 8, [6]). This absence of a coordinate-level holdout violates the assumption of independent and identically distributed (i.i.d.) data, as spatial structure information from the test set is functionally available during training. Under these conditions, the model can degenerate into a location-recognition system, minimizing loss via “spatial look-up” and just memorizing specific geographic backgrounds (e.g., unique driveway configurations) to predict labels. As noted by Kapoor and Narayanan [24], this leakage yields inflated evaluation metrics that reflect the model’s memorization capacity rather than its generalization to unseen geographies [25, 26].

2.3 Corrective Resampling Protocol

To mitigate dataset leakage and improve the validity of generalization estimates under spatial autocorrelation, a spatial coordinate-aware resampling pipeline is implemented. This procedure enforces spatial disjointness by collapsing every geographic coordinate to exactly one retained image and assigning splits at a coarser spatial block level. The correction targets three invariants. First, *Coordinate Disjointness* ensures that the sets of coordinate proxies $\mathcal{C}$ in each split are mutually exclusive ($\mathcal{C}_{train} \cap \mathcal{C}_{val} = \emptyset$, etc.). Second, *Unique Retention* mandates that exactly one sample is kept per coordinate c to eliminate conflicting pre/post pairs. Third, *Reduced Proximity Leakage* requires that nearby coordinates be preferentially co-assigned to the same split to minimize spatial autocorrelation. The resampling process is as follows:

1. **Parsing and Input Scan:** Labels are derived from the immediate directory (*damage* or *no_damage*). Coordinate proxies are parsed from filenames using regex. The algorithm extracts all signed floating-point substrings and interprets the last two values as the coordinate tuple. Therefore, filenames must contain at least two floats (as a proxy for $\approx \langle lon \rangle_ \langle lat \rangle$). Files failing this condition are dropped.
2. **Coordinate-level deduplication ("Keep-One").** For each coordinate group G_c , exactly one image is retained, chosen uniformly at random. This removes exact coordinate duplicates and prevents the same physical location from appearing multiple times across splits. It also removes explicit pre/post pairs at identical coordinates when both classes occur at the same c . However, it does not eliminate the underlying label-time confounding induced by the dataset construction, since *damage* tiles are sourced from post-event imagery. In contrast, *no_damage* tiles are sourced from pre-event imagery at corresponding locations [6]. Consequently, global acquisition and photometric differences can still act as shortcuts unless post-event negatives or matched acquisition conditions are introduced.
3. **Spatial Blocking and Split Assignment:** Coordinates are discretized into unique spatial blocks $b \in \mathcal{B}$ by rounding to a precision of 1 decimal place, creating grid cells of approximately 11.1 km (lat.) $\times$ 9.5 km (lon.). The set of unique blocks $\mathcal{B}$ is randomly partitioned into train (70%), validation (15%), and test (15%) splits. A guard clause ensures that evaluation splits receive at least one block, provided that the total block count $|\mathcal{B}| > 2$. This block-level assignment ignores sample density and class distribution. The resampled splits, therefore, diverge from the target ratios, and the spatial clustering of damage leads to label imbalance across splits.

The resampling reduced the dataset from 23'000 to 14'223 images (-38.2%). The resulting split composition is 12'600 (Train), 495 (Validation), and 1'128 (Test). The impact of the spatial blocking strategy on the geographic distribution of splits is visualized in Appendix Figure A.1. Table 2.1 details the reduction in sample size.

Table 2.1: Dataset Inventory: original (Leaked) vs. Resampled (Corrected). The resampling process reduces the dataset size, particularly in the validation split, to enforce spatial independence.

Metric	Original Dataset	Resampled Dataset
Total Images	23'000	14'223
<i>Damage (Post-event)</i>	15'000	10'529
<i>No Damage (Pre-event)</i>	8'000	3'694
Split Distribution		
<i>Train</i>	19'000	12'600
<i>Validation</i>	2'000	495
<i>Test</i>	2'000	1'128

This strategy was selected over alternative controls. A *Keep-All* strategy (forcing all images at a coordinate to a single split) would preserve sample size while retaining the pre/post pairs that enable "Time Detector" shortcut learning. A *Drop Conflicted* strategy (removing all ambiguous coordinates) avoids the label noise but risks systematically excluding difficult transition zones. The implemented *Keep-One* approach with block splitting provides the most robust guarantee of spatial independence.

3. Data Leakage Analysis

This chapter assesses whether the provided Hurricane Harvey dataset permits solutions via non-causal shortcuts and the extent to which the coordinate-aware resampling mitigates these artifacts.

3.1 Spatial Redundancy and Label Consistency

To quantify the extent of spatial leakage, geospatial coordinates were parsed from filenames and treated as location identifiers. Four specific leakage indicators were computed to detect violations of the independence assumption: *multi-image coordinates* count locations sampled more than once; *label conflicts* identify identical coordinates associated with contradictory pre/post labels; *cross-split overlap* tracks coordinates appearing in multiple splits; and *proximity leakage* quantifies the total count of cross-split pairs located within a 50 m radius based on Haversine distance.



Figure 3.1: Visual demonstration of spatial data leakage resulting from the dataset generation. The figure shows samples generated at the same geographic coordinates ($lon = -95.2698, lat = 29.6052$). The panels illustrate the persistence of background spatial features (e.g., vegetation patterns, road geometry) across different temporal acquisitions. One sample is assigned to the *no_damage* class (training split) and another to the *damage* class (test split).

Analysis reveals severe leakage in the dataset. Using filename-derived coordinate tuples (longitude, latitude) revealed that the original dataset contains 14'223 unique coordinate tuples, of which 7'342 occur in multiple images. 2'972 coordinate tuples occur in more than one split, violating the assumption of independent evaluation (see Figure 3.1). Even when exact-coordinate overlaps are ignored, the split still contains 8'589 train-test and 1'720 train-validation image pairs within a Haversine distance of 50 m. This spatial autocorrelation enables memorization of location (e.g., shared acquisition conditions, illumination, seasonal signatures, and background structure), inflating test performance beyond true generalization. As shown in Appendix Figure A.1, this splitting strategy yields dense, spatially intermingled clusters in which training, validation, and test samples coexist in immediate proximity, thereby failing to enforce spatial independence.

Table 3.1: Leakage indicators before/after resampling. Near-neighbor counts use a threshold of 50 m.

Dataset	Images	Unique coords	Conflicts	Multi-img	Multi-split	Cross-split (≤ 50 m)
original	23'000	14'223	6'834	7'342	2'972	32'078
Resampled	14'223	14'223	0	0	0	29

The resampling protocol eliminates exact-coordinate leakage. The corrected dataset contains one image per coordinate, yielding zero multi-image coordinates, zero coordinate label conflicts, and zero coordinates shared across splits. Proximity-based leakage is strongly reduced but not eliminated. Cross-split near-neighbor pairs within 50 m drop from 32'078 to 29, implying that most cross-split similarity in the original dataset was driven by duplicated or near-duplicated locations rather than unavoidable geographic adjacency.

3.2 Spectral Analysis and the “Time Detector” Hypothesis

Kernel density estimates reveal a spectral shift between classes (Appendix Figure A.3), with the mean global brightness of *damage* tiles exceeding *no_damage* by roughly 12.5 intensity levels on the 8-bit $[0, 255]$ scale. To test whether such photometric differences can drive classification in isolation, a controlled test trains identical baseline CNNs on three variants: **(I)** the original dataset, **(II)** the resampled dataset (according to Section 2.3), and **(III)** a pixel-shuffled version created by randomly shuffling the pixel *positions* within each image using one permutation π that is applied identically to all channels, i.e., the full (R, G, B) triplet at location p is moved to location $\pi^{-1}(p)$ (equivalently $x'_{c,p} = x_{c,\pi(p)}$). This preserves the overall color distribution but destroys spatial structure, such as edges and textures. The model on the original dataset trained reaches an accuracy of 0.873, while the models trained on resampled and shuffled datasets reach 0.765 and 0.762, respectively (Appendix Figure A.5). The near-equivalence of the latter two indicates that global photometric statistics contribute materially to separability. A counterfactual probe isolated the effect of global color/brightness statistics. For each test image, we *re-normalized each RGB channel* so that its mean and standard deviation match those of the opposite class: *damage* images were mapped to the *no_damage* channel-wise (μ, σ) , and vice versa (Appendix Fig. A.7). This changes overall photometry while largely preserving spatial structure. After mapping *damage*→*no_damage* photometry, 417 previously correct *damage* predictions flipped to *no_damage* (35.6% of the test set; Appendix Fig. A.6). The reverse mapping caused only 45 flips. The asymmetry indicates the classifier relies on global photometric/spectral statistics as a decision cue.

3.2.1 Visualizing Shortcut Learning via Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) [27] was used as a final qualitative check. To isolate the effect of the split (and avoid architectural confounds), we applied Grad-CAM to the same input using two instances of the same 4-layer CNN (3×3 convolutions + global average pooling, implemented in *PyTorch*) trained on different datasets only (original vs. resampled). Figure 3.2 shows that the original dataset model yields diffuse attributions that extend into the background context (vegetation, shadows; Fig. 3.2a), whereas the model trained on the resampled dataset concentrates saliency on the building footprint and roof geometry (Fig. 3.2b). This shift is consistent with reduced reliance on context-based shortcuts after data leakage mitigation.

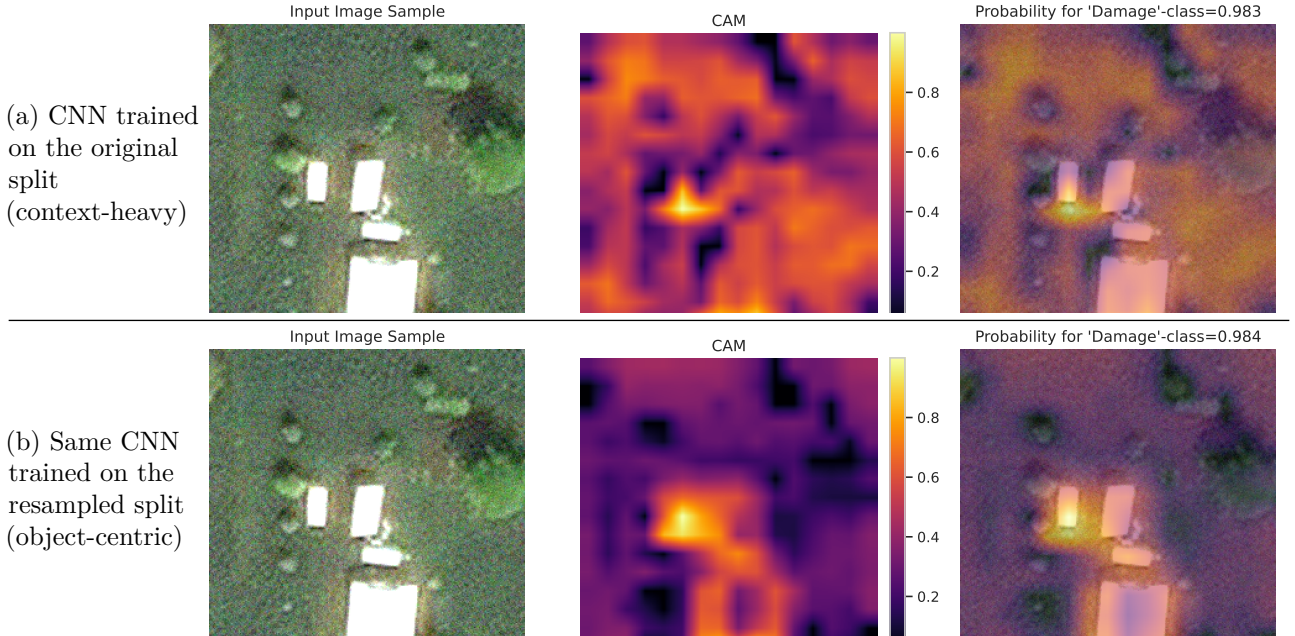


Figure 3.2: Grad-CAM heatmaps for the same test image using identical CNN architectures trained on different splits. Warmer colors indicate higher contribution to the predicted class. The original-split model highlights broad background regions (context cues), while the resampled-split model concentrates on the building footprint/roof (structural cues), consistent with reduced leakage-driven shortcut learning.

4. Model Training and Evaluation

All trainings follow a common training and evaluation pipeline. Each model is trained independently on both the original and the resampled dataset variant on a single NVIDIA RTX 4090 GPU.

4.1 Preprocessing and Selected Model

Images are loaded from the folder structure, and class imbalance is addressed using a *WeightedRandomSampler* [28] that samples with replacement using inverse-frequency weights ($w_{y_i} = N/n_{y_i}$). Preprocessing enforces a split-consistent order: deterministic resizing to a model-specific resolution, tensor conversion, and normalization using Z-score statistics (μ, σ) calculated on the training split. During training, stochastic augmentation is conditionally applied: random horizontal flips, in-plane rotation sampled uniformly from $[-d, d]$, and photometric jitter that independently perturbs brightness, contrast, saturation, and hue. The benchmark comprises seven CNN backbones (ResNet-18/34/50, EfficientNet-B1/B2, ConvNeXt-Tiny/Small) and a *custom CNN* implemented in *PyTorch*. All models are trained under a shared baseline configuration of 50 epochs, AdamW, automatic mixed precision enabled, gradient-norm clipping at 1.0, early-stopping parameter set to 16, label smoothing set to 0.05, and baseline weight decay of 10^{-4} , with additional per-model overrides summarized in Table 4.1.

Table 4.1: Model zoo hyperparameter overrides. Hyperparameters not listed are inherited from the shared PyTorch baseline configuration.

Model	Input size	Batch size	Learning rate	Dropout	Weight decay	Pretrained
ResNet-18	256	64	$3 \cdot 10^{-4}$	0.20	10^{-4}	True
ResNet-34	256	48	$3 \cdot 10^{-4}$	0.20	10^{-4}	True
ResNet-50	256	32	$2.5 \cdot 10^{-4}$	0.25	10^{-4}	True
EfficientNet-B1	240	48	$3 \cdot 10^{-4}$	0.30	10^{-4}	True
EfficientNet-B2	260	36	$3 \cdot 10^{-4}$	0.30	10^{-4}	True
ConvNeXt-Tiny	256	48	$5 \cdot 10^{-4}$	default	$5 \cdot 10^{-5}$	True
ConvNeXt-Small	256	32	$5 \cdot 10^{-4}$	default	$5 \cdot 10^{-5}$	True
Custom CNN	256	96	$1 \cdot 10^{-3}$	0.30	10^{-4}	False

4.2 Evaluation Metrics

Evaluation is performed by a common `evaluate()` routine that runs the model in `eval()` mode under `@torch.no_grad()`, iterates the dataloader, supports batches of either `(images, labels)` or `(images, labels, meta)`, concatenates logits and labels across batches, and converts logits to probabilities using `softmax`. Discrimination metrics are computed from argmax predictions using sklearn: accuracy and macro-precision, macro-recall, and macro-F1 with `average="macro"` and `zero_division=0`. The routine returns both the metric dictionary and the full per-sample outputs (labels, logits, probabilities, predictions), enabling downstream reliability and calibration analysis without recomputation.

4.3 Calibration and Reliability Metrics

Reliability was quantified using the Brier score [29] and ECE [17]. The Brier score is computed as the mean squared error between the predicted probability $p_{i,1}$ and the binary label y_i :

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N (p_{i,1} - \mathbb{1}[y_i = 1])^2. \quad (4.1)$$

ECE is computed using fixed-bin partitioning with $B = 15$ equal-width bins over $[0, 1]$. For each bin b , the absolute gap between empirical accuracy $\text{acc}(b)$ and mean confidence $\text{conf}(b)$ is aggregated. The per-sample confidence is defined as $\hat{c}_i = \max_k p_{i,k}$ and correctness as $\mathbb{1}[\hat{y}_i = y_i]$ with $\hat{y}_i = \arg \max_k p_{i,k}$:

$$\text{ECE} = \sum_{b=1}^B \frac{n_b}{N} |\text{acc}(b) - \text{conf}(b)|. \quad (4.2)$$

This estimate is purely empirical; no smoothing is applied (see review of Wang [16] for alternatives and broader calibration diagnostics).

To correct confidence estimates, post-hoc Temperature Scaling [30] was applied per trained model. A single scalar $T > 0$ is fitted on validation logits by minimizing cross-entropy, producing calibrated probabilities $p_i^{(T)} = \text{softmax}(z_i/T)$. Values $T < 1$ sharpen the distribution (indicating underconfidence on the validation split), while $T > 1$ softens it (indicating overconfidence). Appendix Tables A.2 and A.3 report both the uncalibrated ECE and the post-calibration $\text{ECE}(T)$ together with the fitted T for each model, enabling a direct assessment of whether calibration fitted on validation transfers to the test split.

In emergency-response, these calibration diagnostics operationalize “trustworthy confidence”: a model that achieves high Macro- $F1$ but remains miscalibrated can be operationally harmful because its probabilities cannot be interpreted as decision-relevant risk estimates [30].

4.4 Training Results and Analysis

This section interprets discrimination and reliability jointly, because raw accuracy or Macro- $F1$ can be high while confidence remains systematically mis-scaled. Detailed per-model metrics are provided in Appendix Tables A.2 (original dataset) and A.3 (resampled dataset).

On the **original dataset** (Table A.2), ConvNeXt-Tiny achieves the best test performance (Test Macro- $F1$ /Acc. 0.990/0.990, Brier 0.009), followed closely by ConvNeXt-Small (0.985), ResNet-18 (0.983), and ResNet-34 (0.982), i.e., all within $\Delta F1 \leq 0.008$ of the leader. For ResNet-18 and ConvNeXt-Small, the best epoch occurs at 2 and 4 epochs (Figure A.8), whereas ConvNeXt-Tiny peaks later (best epoch 15; Table A.2). Early selection for some models is compatible with the presence of easy-to-learn cues (including leakage-aligned shortcuts). Still, it is not decisive evidence. The strong epoch-to-epoch fluctuations in Figure A.8 could also indicate optimization sensitivity and/or stochastic validation estimation under the training pipeline. The Custom CNN also performs strongly ($F1$ 0.951, Acc 0.952), supporting the hypothesis that low-level cues exist in this dataset, but these effects are non-uniform: ResNet-50 (0.668) and EfficientNet-B1 (0.675) underperform, with EfficientNet-B2 intermediate (0.745), weakening any claim of universal leakage benefit and instead implicating interactions with optimization overrides (Table 4.1) and potentially different, unauditable initialization. Despite high accuracy, confidence is still mis-scaled (ECE 0.025–0.037). Validation-fitted temperature scaling transfers for strong models (ConvNeXt-Tiny 0.030 $\rightarrow$ 0.013, ResNet-18 0.025 $\rightarrow$ 0.012, ConvNeXt-Small 0.037 $\rightarrow$ 0.018, Custom CNN 0.056 $\rightarrow$ 0.024), whereas weaker models remain poorly calibrated after scaling (EfficientNet-B1 0.208 $\rightarrow$ 0.185, ResNet-50 0.145 $\rightarrow$ 0.131) and retain high Brier scores (0.199–0.230), indicating primary failure by misclassification rather than a simple calibration offset.

On the **resampled dataset** (Table A.3), a performance inversion occurs. ResNet-50 becomes best (Test Macro- $F1$ 0.891, Acc 0.922, Best Epoch 14), followed by ResNet-34 (0.885, 0.918, Best Epoch 12), while ConvNeXt-Tiny drops (0.990 $\rightarrow$ 0.862) and ConvNeXt-Small drops (0.985 $\rightarrow$ 0.823). The shifts are large (ResNet-50 +0.223: 0.668 $\rightarrow$ 0.891; EfficientNet-B1 +0.186: 0.675 $\rightarrow$ 0.861; Custom CNN −0.156: 0.951 $\rightarrow$ 0.795). This is consistent with reduced shortcut availability forcing reliance on more complex representations, but remains confounded by nonidentical optimization settings (batch size, learning rate, resolution) and unauditable initialization; the defensible claim is therefore that resampling changes which architecture–optimization combinations generalize best, making selection on the original split directionally risky. Reliability also shifts: uncalibrated ECE is low for the top ResNets (ResNet-50 0.029, ResNet-34 0.031) but moderate elsewhere (ConvNeXt-Small 0.074, EfficientNet-B2 0.089). At the same time, Brier scores increase relative to near-ceiling models (e.g., ResNet-50 0.061, ResNet-34 0.065) due to higher error and uncertainty. Crucially, validation-fitted temperature scaling fails to transfer, with $\text{ECE}(T)$ consistently worse than ECE (e.g., ResNet-50 0.029 $\rightarrow$ 0.036, ResNet-34 0.031 $\rightarrow$ 0.038, EfficientNet-B1 0.050 $\rightarrow$ 0.063), implying calibration overfitting and/or residual distribution shift plus small-sample noise; default temperature scaling is therefore unjustified without safeguards such as cross-validation or larger calibration sets.

4.5 Hyperparameter Tuning of ResNet-34

Although ResNet-50 achieved the highest nominal performance on the resampled dataset ($F1 = 0.891$), ResNet-34 followed closely ($F1 = 0.885$). Given the statistically marginal gap ($\Delta \approx 0.006$), **ResNet-34** was selected for optimization to prioritize computational efficiency. Bayesian optimization was performed via Optuna (TPE sampler, $N_{\text{startup}} = 10$, 35 trials) to maximize validation Macro-F1. The search space covered: learning rate $[5 \times 10^{-5}, 5 \times 10^{-3}]$ (log-uniform), weight decay $[10^{-6}, 5 \times 10^{-4}]$ (log-uniform), dropout $[0, 0.35]$, batch size $\{24, \dots, 56\}$, and resolution $\{224, \dots, 288\}$. The core configuration (AdamW, Automatic Mixed Precision, clipping, label smoothing) was retained, with early stopping tightened to 8 epochs to limit overfitting. Hyperparameter importance analysis (Appendix Figure A.9) confirms that learning rate dominates, followed by weight decay and dropout, while throughput parameters (batch/image size) are negligible. This indicates an overfitting-limited regime where generalization is governed by regularization strength rather than capacity, further supported by the partial dependence plots (Appendix Figure A.10). However, the tuned configuration (Accuracy 0.909, F1 0.867) failed to surpass the fixed baseline (Accuracy 0.918, F1 0.885). This shortfall suggests validation overfitting or an insufficient search budget; indeed, the validation loss trajectory (Appendix Figure A.11) diverges after the initial plateau, indicating that prolonged training risks overfitting to the deduplicated split. Errors remain dominated by false positives on *no_damage* (Appendix Figure A.12), reflecting residual domain shifts and highlighting that symmetric macro metrics may mask thresholding issues critical for asymmetric downstream risks.

5. Limitations and Conclusion

Resampling reduces leakage but does not eliminate all confounders. The dataset is temporally confounded by construction, with *no_damage* drawn from pre-event imagery and *damage* from post-event imagery, so the Resampled Dataset mainly suppresses spatial shortcuts while leaving label-aligned acquisition artifacts (illumination, wet ground, haze, debris field context) partially intact; the pixel-shuffle probe (accuracy 0.762) and the asymmetric profile-transfer flips (417 vs 45) are consistent with a strong photometric channel (Figure A.5, Figure A.6). This does not prove that models ignore morphology. Still, it implies that a model trained on this dataset may be biased toward predicting *damage* in post-event scenes, even when structures are intact, which is incompatible with operational triage without additional controls. The resampling also shifts the class distribution (original: 15'000/23'000 $\approx$ 65.2% positive, Table 2.1), altering the operating point of accuracy and calibration-related metrics and complicating direct comparability unless prevalence-robust metrics are also reported. The validation split is small and highly imbalanced ($N = 495$, 88.1% positive), inflating selection variance, weakening hyperparameter conclusions, and plausibly contributing to calibration instability.

To break temporal confounding, the dataset would require *post-event negatives* (post-hurricane imagery of intact structures) or matched acquisition conditions across classes. Without such controls, separating structural damage cues from environment and acquisition artifacts is not identifiable without strong assumptions. To quantify selection uncertainty, spatial k -fold cross-validation over blocks should replace a single fixed split, converting point estimates into confidence intervals at a computational cost of $\approx k \times$, repeated runs across multiple random seeds are additionally required wherever the pipeline is stochastic (conflict tie-breaks, sampling with replacement, augmentations). Calibration should be treated as a domain-generalization problem: the observed degradation of ECE after temperature scaling on the Resampled Dataset (Table A.3) is consistent with geographically distinct domains (e.g., housing types, land cover, industrial vs residential) that induce different confidence distributions, so a single scalar T fitted on one block can overfit and harm calibration elsewhere. Cross-fitted calibration and larger calibration sets are prerequisites before applying temperature scaling by default. Uncertainty quantification should move beyond scalar rescaling and use, e.g., deep ensembles, which can provide more robust uncertainty under shift than single-model softmax confidence, at additional training and inference cost. Also, external validity should be evaluated cross-event with models trained on Harvey should be tested without fine-tuning on later hurricanes (e.g., Ida, Ian) and under controlled sensor/season shifts.

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A. Appendix

Tables

Table A.1: Runtime summary (wall-clock) measured during training on one NVIDIA RTX 4090. Throughput is approximated as train-set size divided by mean epoch time.

Dataset	Model	Epochs	Mean epoch (s)	Total (min)	Throughput (img/s)
Original Dataset	convnext_small	21	25.28	9.29	752
Original Dataset	convnext_tiny	50	17.13	14.57	1109
Original Dataset	custom_cnn	50	5.69	4.84	3339
Original Dataset	efficientnet_b1	37	11.53	7.41	1648
Original Dataset	efficientnet_b2	50	12.36	10.27	1537
Original Dataset	resnet18	50	7.49	6.36	2537
Original Dataset	resnet34	34	11.10	6.44	1711
Original Dataset	resnet50	50	14.12	11.77	1345
Resampled Dataset	convnext_small	50	18.58	15.05	678
Resampled Dataset	convnext_tiny	50	12.96	10.55	972
Resampled Dataset	custom_cnn	50	4.39	3.71	2870
Resampled Dataset	efficientnet_b1	23	7.64	2.99	1649
Resampled Dataset	efficientnet_b2	38	8.25	5.31	1527
Resampled Dataset	resnet18	50	5.17	4.42	2437
Resampled Dataset	resnet34	50	7.18	6.18	1755
Resampled Dataset	resnet50	50	8.48	7.23	1486

Table A.2: Detailed test set performance on the **Original Dataset**. T indicates the optimal temperature parameter fitted on the validation set.

Model	Val Macro- $F1$	Best ep.	Test Macro- $F1$	Acc.	Macro-Prec.	Macro-Rec.	Brier	ECE	ECE(T)	T
ConvNeXt-Tiny	0.990	15	0.990	0.990	0.990	0.990	0.009	0.030	0.013	0.817
ConvNeXt-Small	0.986	4	0.985	0.985	0.985	0.985	0.014	0.037	0.018	0.798
ResNet-18	0.985	2	0.983	0.983	0.983	0.983	0.014	0.025	0.012	0.841
ResNet-34	0.981	7	0.982	0.982	0.982	0.982	0.016	0.030	0.015	0.829
Custom CNN	0.958	35	0.951	0.952	0.952	0.952	0.037	0.056	0.024	0.700
EfficientNet-B2	0.743	28	0.745	0.756	0.764	0.755	0.189	0.140	0.116	1.163
EfficientNet-B1	0.668	5	0.675	0.703	0.735	0.703	0.230	0.208	0.185	1.201
ResNet-50	0.675	4	0.668	0.698	0.739	0.698	0.199	0.145	0.131	1.129

Table A.3: Detailed test set performance on the **Resampled Dataset** under the fixed default configuration (before dedicated hyperparameter tuning). Note the performance inversion for ResNet-50 compared to Table A.2.

Model	Val Macro- $F1$	Best ep.	Test Macro- $F1$	Acc.	Macro-Prec.	Macro-Rec.	Brier	ECE	ECE(T)	T
ResNet-50	0.975	14	0.891	0.922	0.929	0.870	0.061	0.029	0.036	0.925
ResNet-34	0.961	12	0.885	0.918	0.927	0.867	0.065	0.031	0.038	0.924
ConvNeXt-Tiny	0.953	6	0.862	0.902	0.916	0.838	0.082	0.052	0.061	0.916
EfficientNet-B1	0.967	26	0.861	0.906	0.923	0.827	0.080	0.050	0.063	0.864
ResNet-18	0.935	15	0.857	0.903	0.920	0.823	0.076	0.037	0.044	0.933
ConvNeXt-Small	0.914	45	0.823	0.876	0.899	0.795	0.105	0.074	0.079	0.944
Custom CNN	0.896	27	0.795	0.849	0.863	0.757	0.123	0.056	0.058	0.967
EfficientNet-B2	0.870	32	0.775	0.857	0.881	0.727	0.132	0.089	0.095	0.942

Supplementary Figures

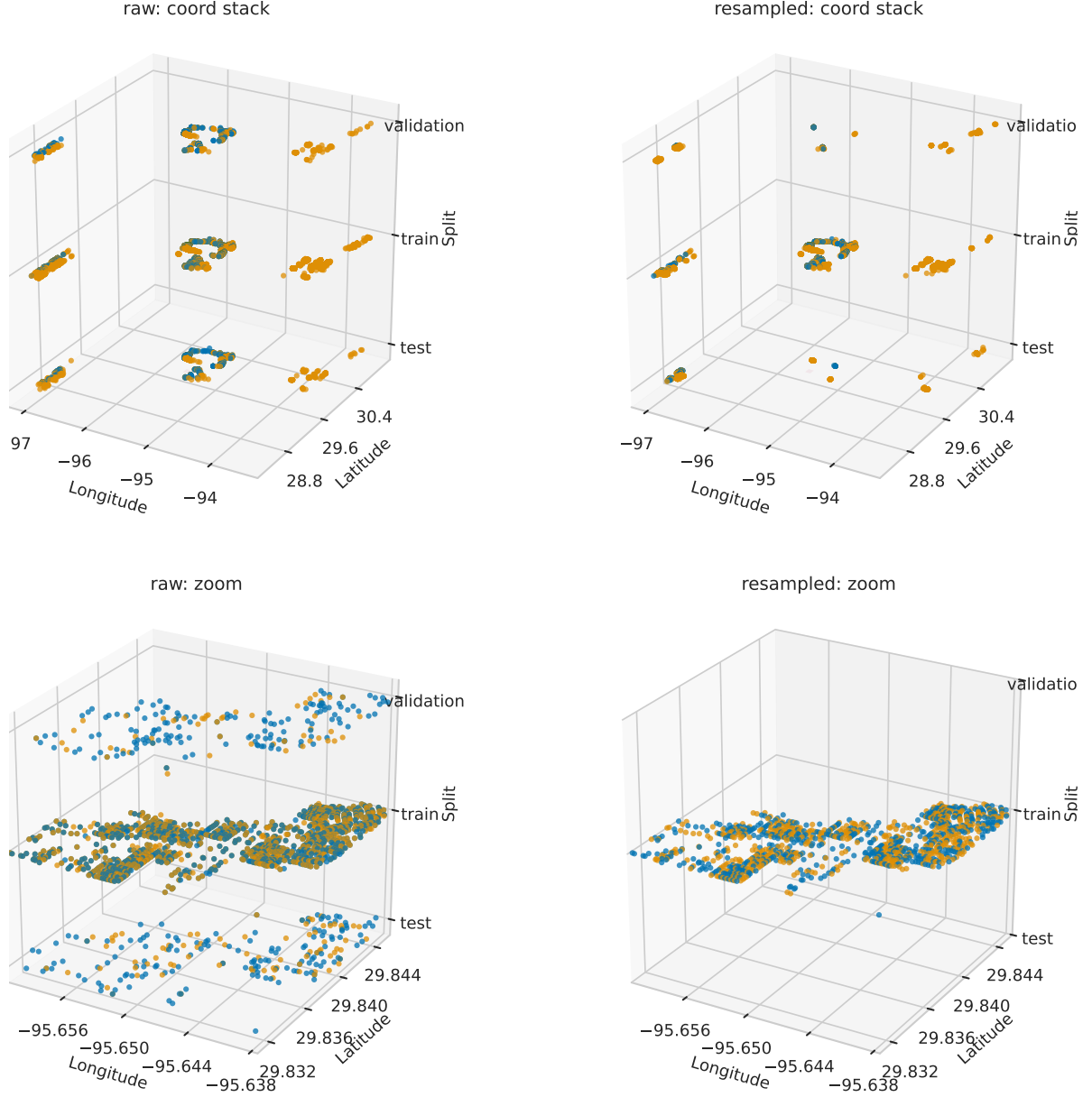


Figure A.1: Original versus leakage-reduced dataset structure visualized as a 3D stacked geospatial scatter plot. Each marker corresponds to a single image tile, whose geographic coordinates (x-axis: longitude, y-axis: latitude) are used to position it. The z-axis is a categorical stacking of the dataset splits (test, train, validation) to make cross-split overlap visible; orange markers denote *damage*, blue markers denote *no_damage* samples. **Top row (coord stack, full extent):** Left shows the original dataset, where train/validation/test tiles are densely intermingled across the same spatial regions. Vertical alignments of markers at identical (lon, lat) across multiple z-levels indicate coordinate reuse across splits (coordinate-level leakage). The right panel shows the resampled dataset, in which coordinate-level deduplication removes most vertical stacks, and a block-based assignment groups nearby coordinates into the same split. **Bottom row (zoomed window):** A magnified region highlights the local neighborhood structure. In the original split (left), small spatial clusters are repeatedly cut across splits, making near-neighbour leakage likely. In the resampled split (right), local neighborhoods are largely retained within a single split, reducing both coordinate reuse and short-range proximity leakage. The block size is on the order of 0.1° (roughly 10 to 11 km), which increases separation but cannot guarantee full independence under long-range spatial autocorrelation or when correlations extend beyond the chosen block scale.

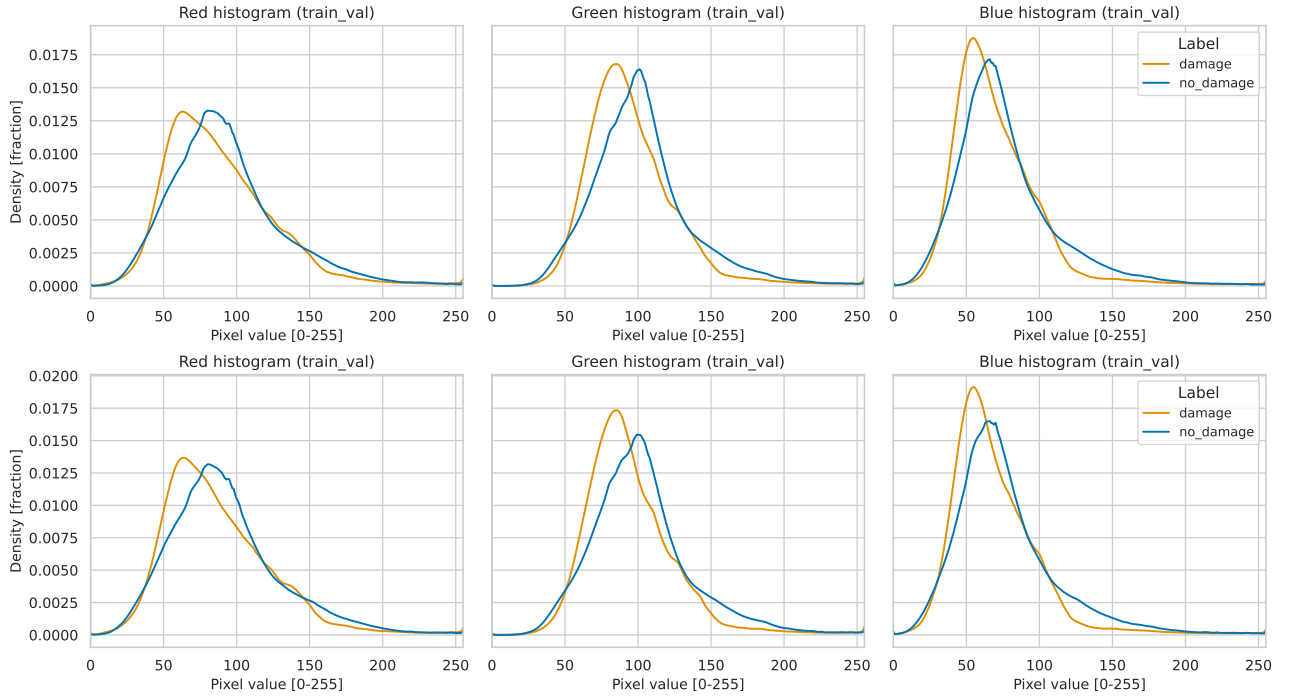


Figure A.2: Channel-wise pixel-intensity distributions (kernel density estimates) for RGB, stratified by label, calculated on train and validation split. **Top:** original dataset. **Bottom:** resampled dataset. In both datasets, the *damage* class is shifted toward lower intensities across channels relative to the *no_damage* class, indicating a label-aligned photometric domain shift that persists after resampling.

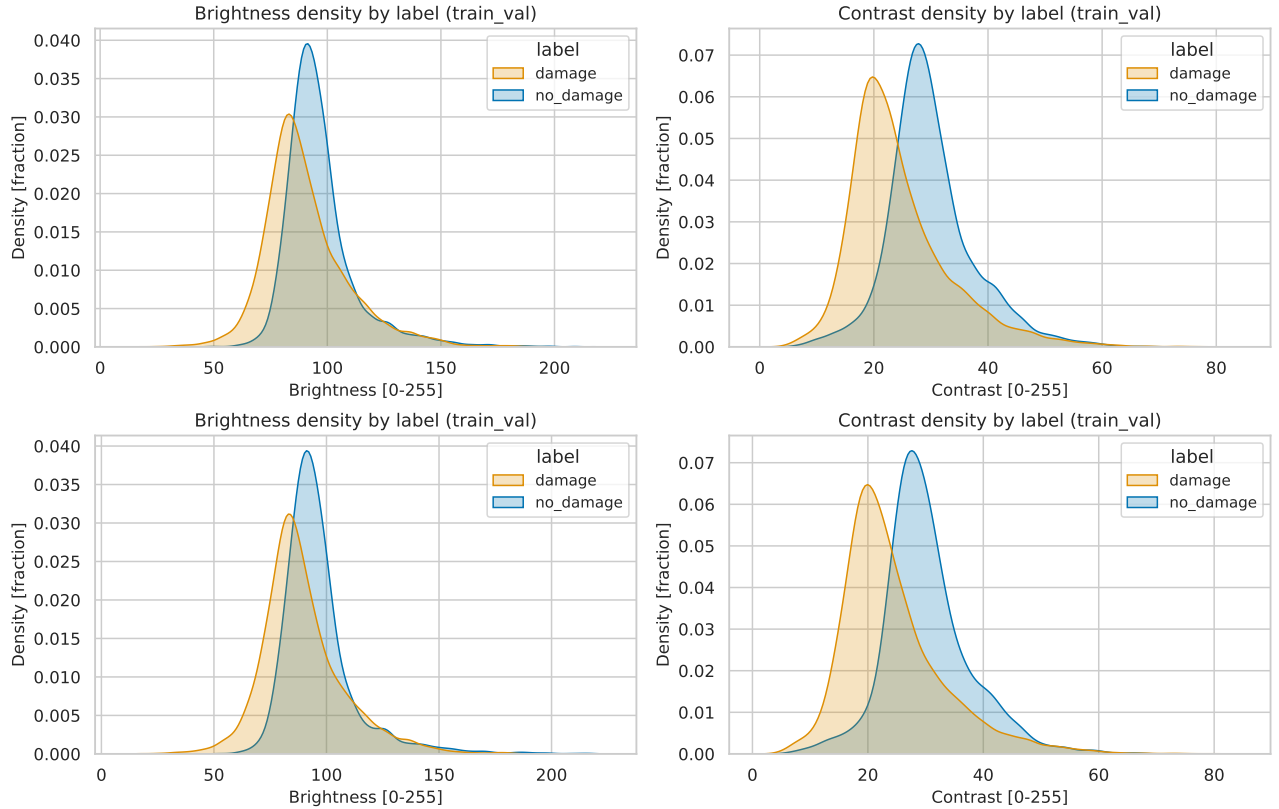


Figure A.3: Global photometric statistics (kernel density estimates) for mean brightness and contrast, calculated on train and validation split. **Top:** original dataset. **Bottom:** resampled dataset. The *damage* class concentrates at lower brightness and lower contrast than *no_damage* in both versions, consistent with a label-aligned acquisition/photometric shift. Resampling removes coordinate duplicates and changes split assignment but does not remove photometric domain shift.



Figure A.4: Visual demonstration of the spatial structure ablation used to test for shortcut learning. Left: Original input image from the dataset. Right: The same image after applying the *PixelShuffleDestroyer* transformation. By destroying all geometric features (edges, shapes, textures) while preserving the exact global color histogram and spectral statistics, this transformation isolates the model’s reliance on domain-specific environmental cues (e.g., lighting, haze) versus structural damage features.

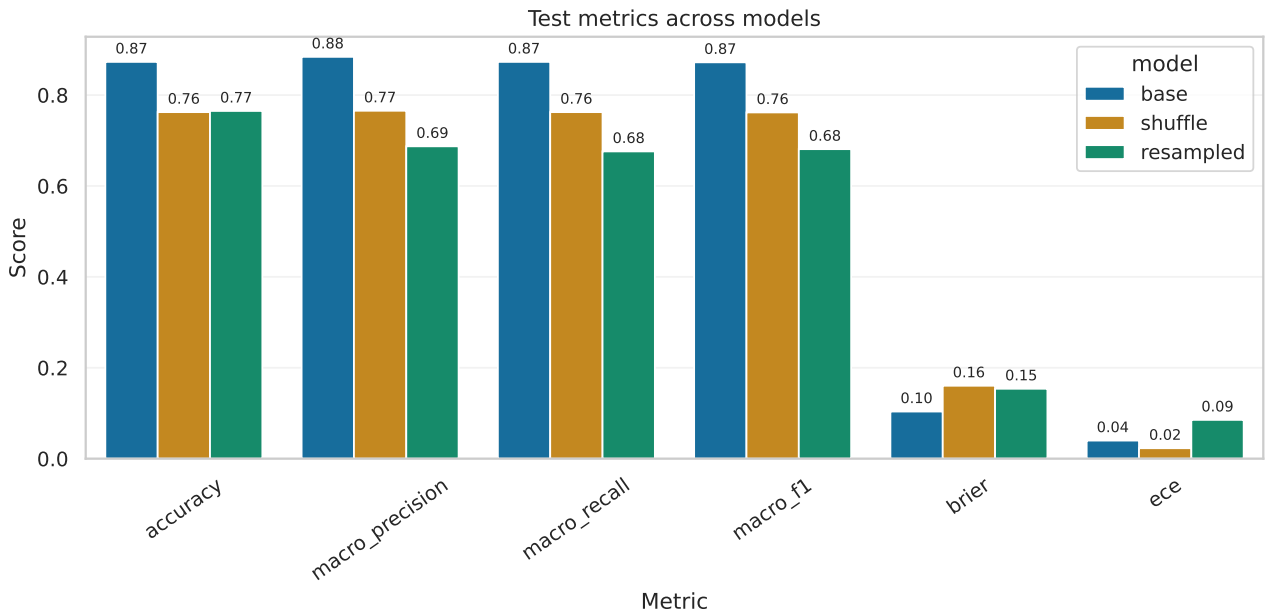


Figure A.5: Controlled leakage probe (fixed baseline CNN): comparison of metrics across the original, the resampled, and a pixel-shuffled dataset variant that preserves per-channel histograms while destroying spatial structure. The model trained on the original dataset performs best within this probe. The near match between resampled and shuffled performance indicates a substantial contribution of global photometric cues.

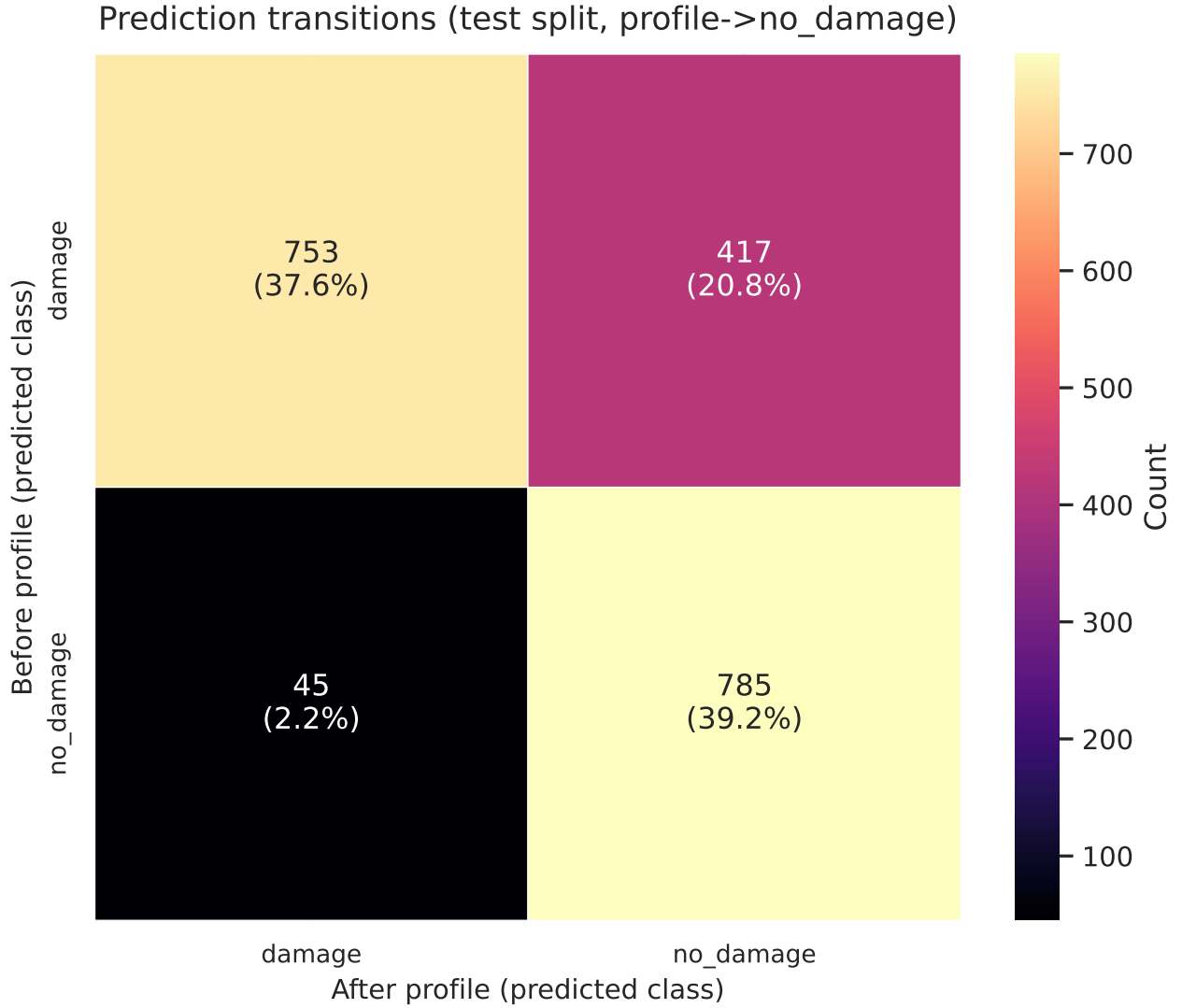
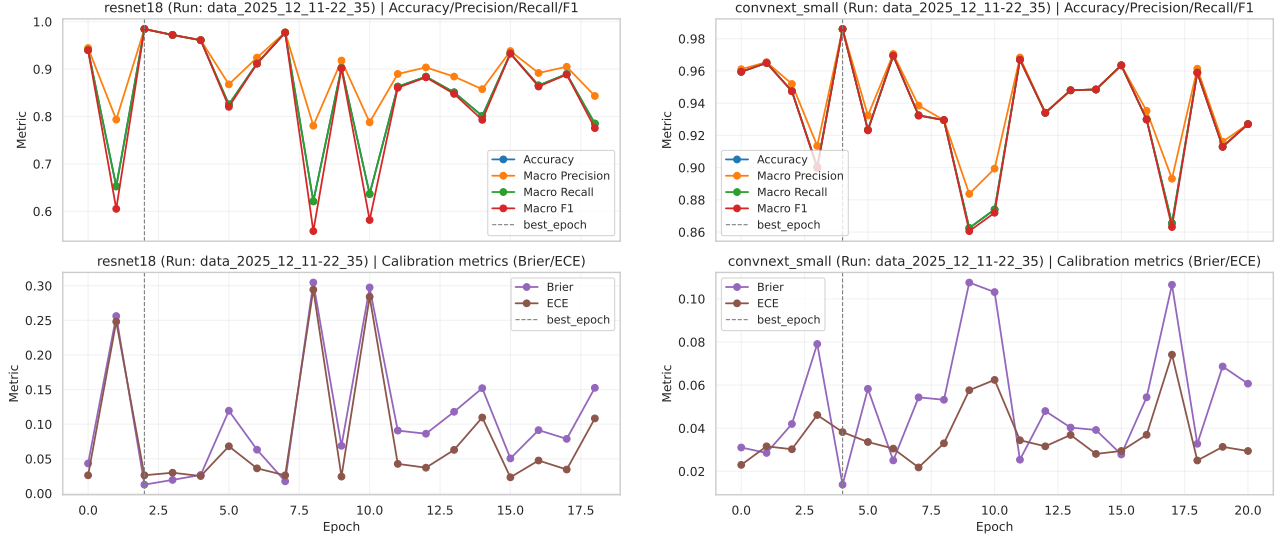


Figure A.6: Transition matrix of model predictions under photometric intervention. The vertical axis represents the predicted class for original images, while the horizontal axis represents the prediction after the image’s color profile is shifted to match the *no_damage* domain. The intervention causes a significant mode collapse: 417 samples (representing $\approx 35.6\%$ of the images originally classified as *damage*) are flipped to *no_damage* solely due to changes in lighting and atmospheric statistics. In contrast, predictions for the *no_damage* class remain largely stable (785 unchanged vs. 45 flips), indicating the model’s specific sensitivity to the “pre-event” spectral signature.



Figure A.7: Qualitative demonstration of the counterfactual photometric probe. The left column displays original samples, while the right column shows the same images with their global color statistics (brightness, contrast, and channel means) adjusted to match the distribution of the *no_damage* class (pre-event domain). The model's predicted probability for the *damage* class (p_{damage}) consistently decreases after this intervention (e.g., from 0.99 to 0.72), indicating that the model relies on global environmental cues rather than purely structural features to support its damage predictions.



(a) ResNet-18: validation discrimination and calibration metrics per epoch on the original split. The dashed line marks the selected best epoch based on the Macro F1 score (2).

(b) ConvNeXt-Small: validation discrimination and calibration metrics per epoch on the original split. The dashed line marks the selected best epoch based on the Macro F1 score (4).

Figure A.8: Training trajectories used for early-stopping model selection on the original dataset (metrics shown per epoch; dashed line denotes the selected checkpoint).

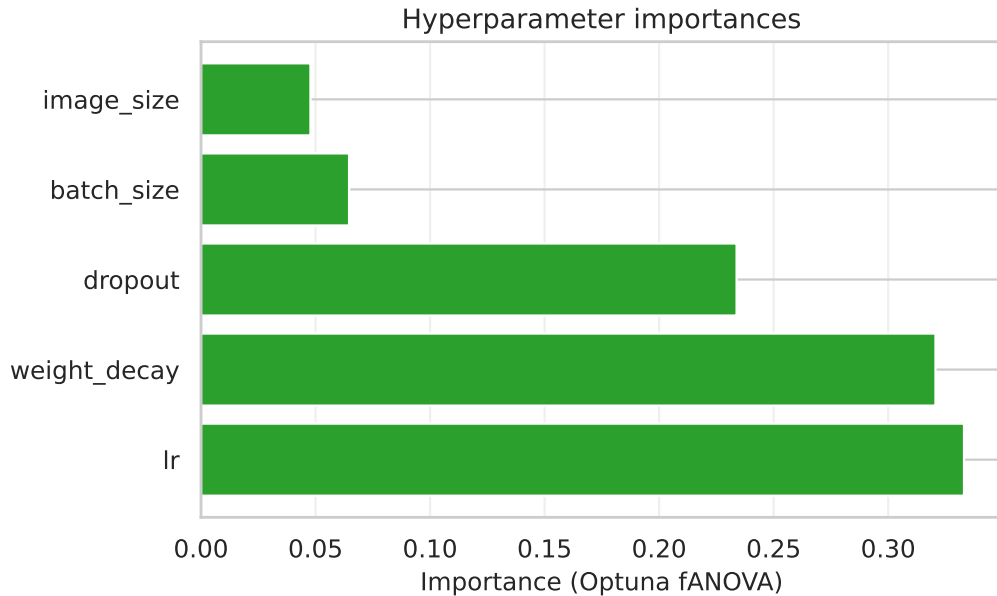
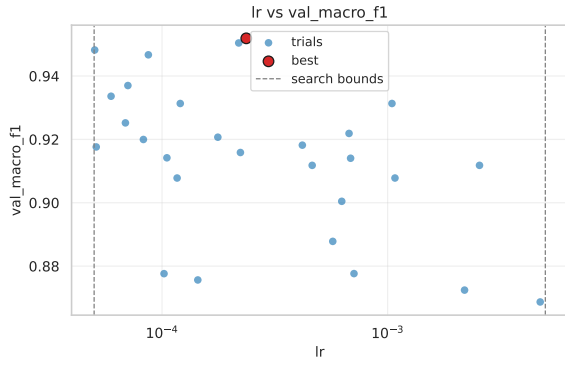
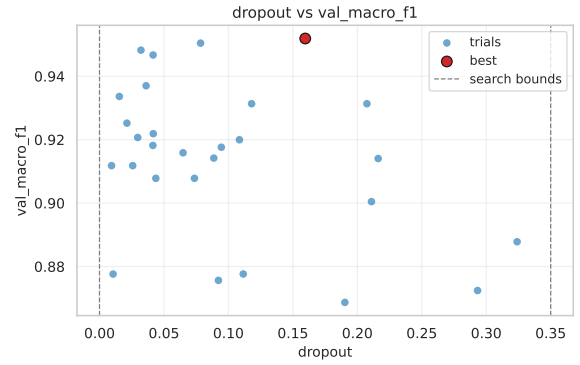


Figure A.9: Hyperparameter importance derived from the Bayesian optimization study (fANOVA). Learning rate and weight decay dominate model performance, confirming that regularization is the critical lever for generalization on the resampled dataset.



(a) Learning Rate vs. Val F1



(b) Dropout vs. Val F1

Figure A.10: Partial dependence plots from the hyperparameter search. The red line indicates the best trial. Performance peaks at lower learning rates (10^{-4}) and moderate dropout (0.15).

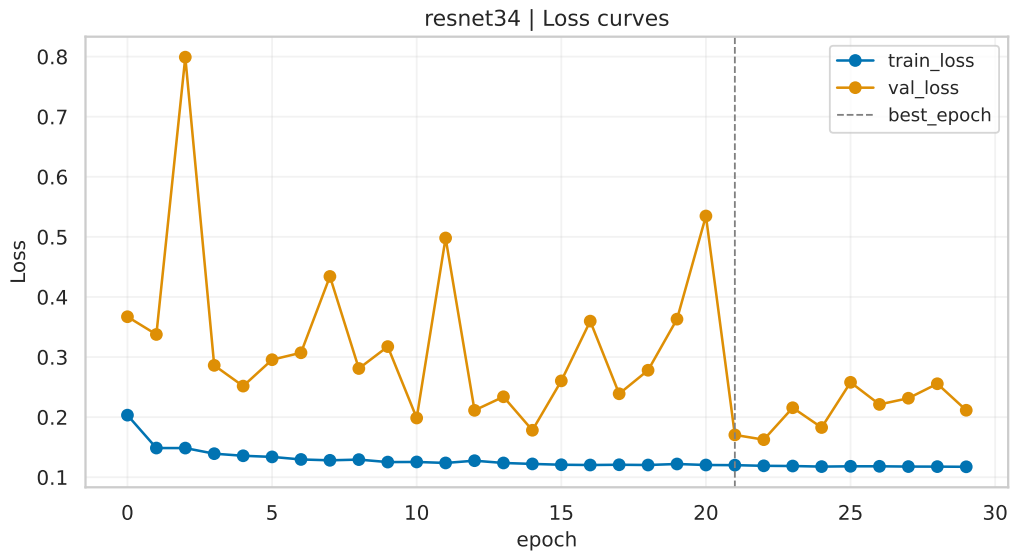


Figure A.11: Training and validation loss curves for the optimal ResNet-34 model. The divergence of validation loss after epoch 12 illustrates the overfitting risk inherent to the small dataset size, necessitating early stopping.

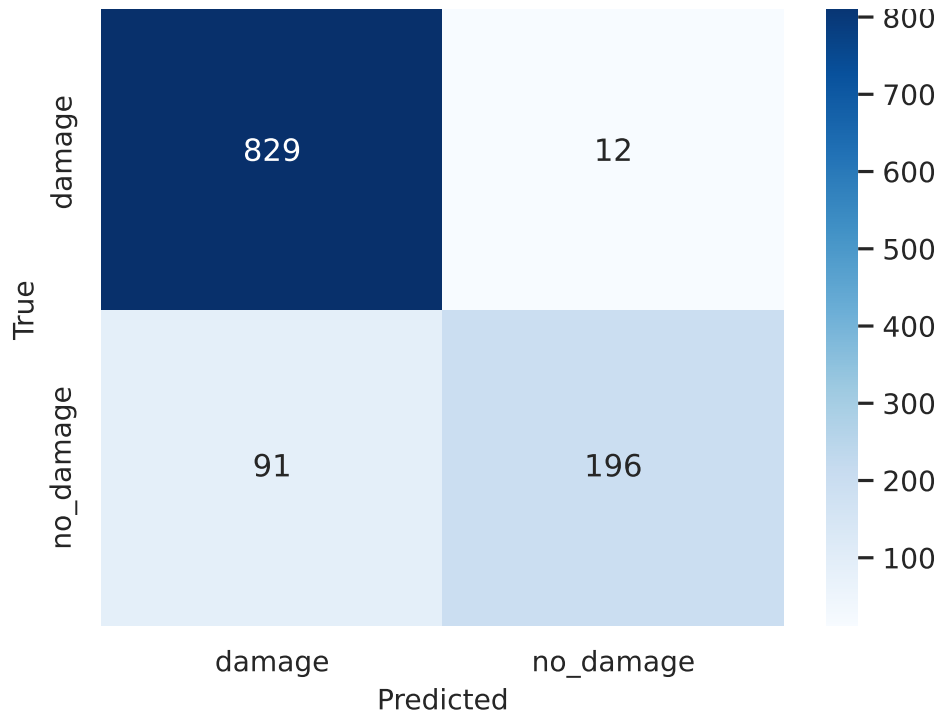


Figure A.12: Confusion matrix for the optimal ResNet-34 model evaluated on the leakage-corrected test set. Errors are dominated by false positives ($FP = 91$) relative to false negatives ($FN = 12$), indicating a bias toward predicting *damage* on intact samples, consistent with residual domain shift.

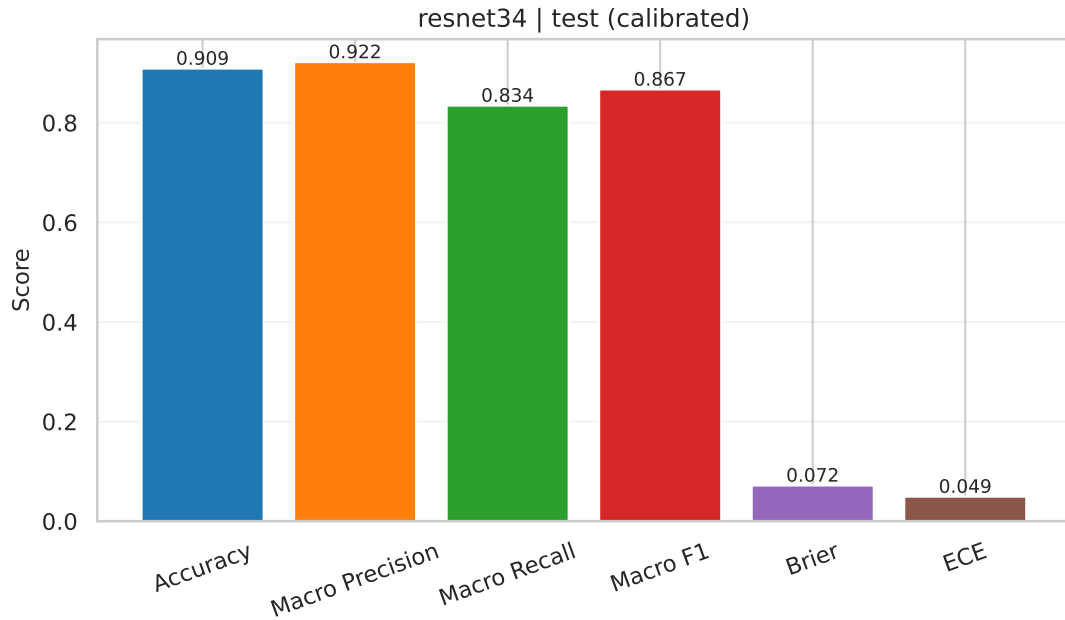


Figure A.13: Aggregate performance metrics on the test set following post-hoc calibration. The model achieves high precision (0.922) but lower recall (0.834), resulting in a Macro-F1 of 0.867. Reliability metrics (Brier score 0.072, ECE 0.049) indicate moderate calibration error despite temperature scaling.

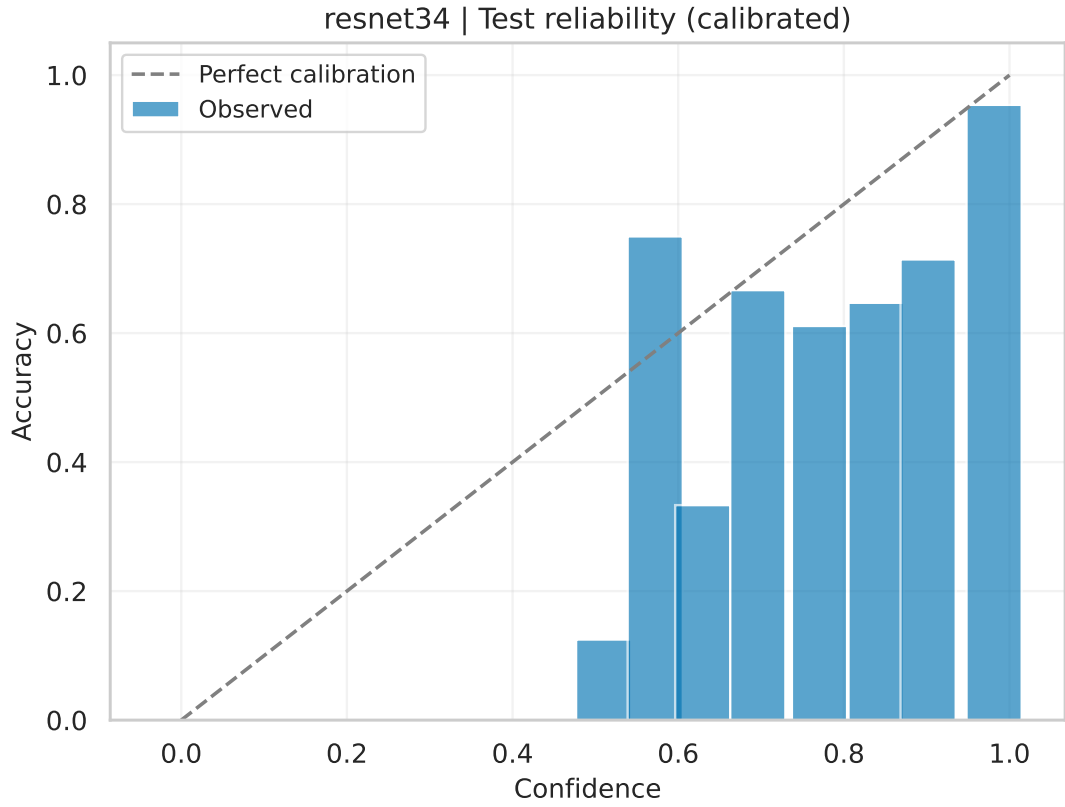


Figure A.14: Reliability diagram for the calibrated ResNet-34 model on the test set. The curve tracks the mean predicted confidence against empirical accuracy within 15 fixed-width bins. Deviations from the diagonal (perfect calibration) reveal that the model remains slightly underconfident in the mid-range probability bins (0.4–0.6) even after temperature scaling.

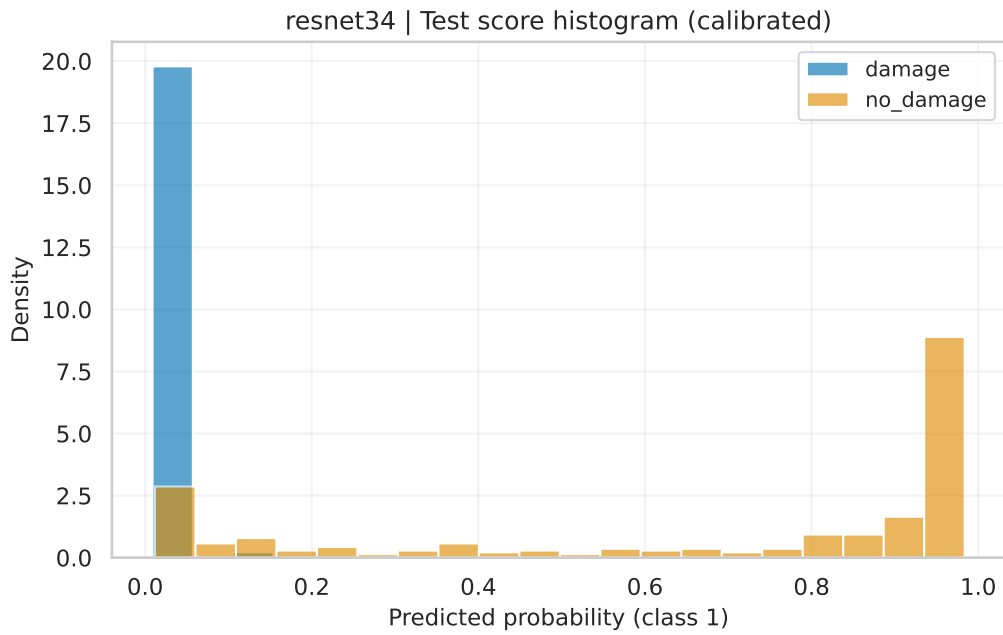


Figure A.15: Calibrated score distribution on the test set. Despite temperature scaling ($T = 0.925$), predictions remain highly polarized, indicating that the model retains strong confidence in its classifications.